Building the Digital Site Manager

A No-Code Revolution for Construction Knowledge

MICROSOFT SHAREPOINT · POWER AUTOMATE · AI

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The Problem: Information Silos

In construction, the gap between a successful project and a costly delay often comes down to **information accessibility**. Critical safety protocols, site setup manuals, and historical project data are buried in thousands of PDFs and Word documents across scattered folders.

⚠ When a site engineer needs an answer, they shouldn't spend hours digging through a digital filing cabinet.

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The Solution: Three Tools, One Ecosystem

SharePoint

Your unified digital library — the "brain" storing all corporate memory.

Power Automate

Your digital courier — the "hands and feet" that trigger workflows automatically.

AI (GPT-4)

Your experienced project manager — reads all past files and answers instantly.

No team of programmers required. Any construction firm can build this using tools they likely already own.

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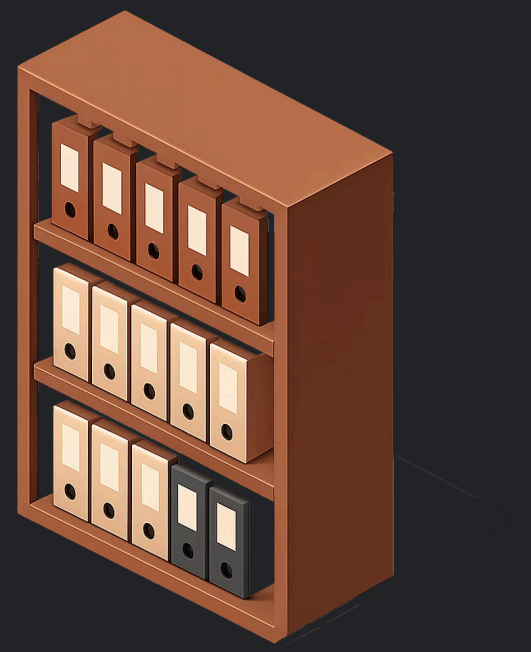
Foundation: SharePoint as Your Project Hub



For most firms, SharePoint is already part of Microsoft 365. Instead of a "dumping ground," treat it as your **corporate memory**.

- Centralize all historical project data, manuals, and vendor contracts
- Tag every file with **Metadata** — "Project Type," "Location," "Safety Level"
- Helps AI identify the most relevant **Board-Approved** document version

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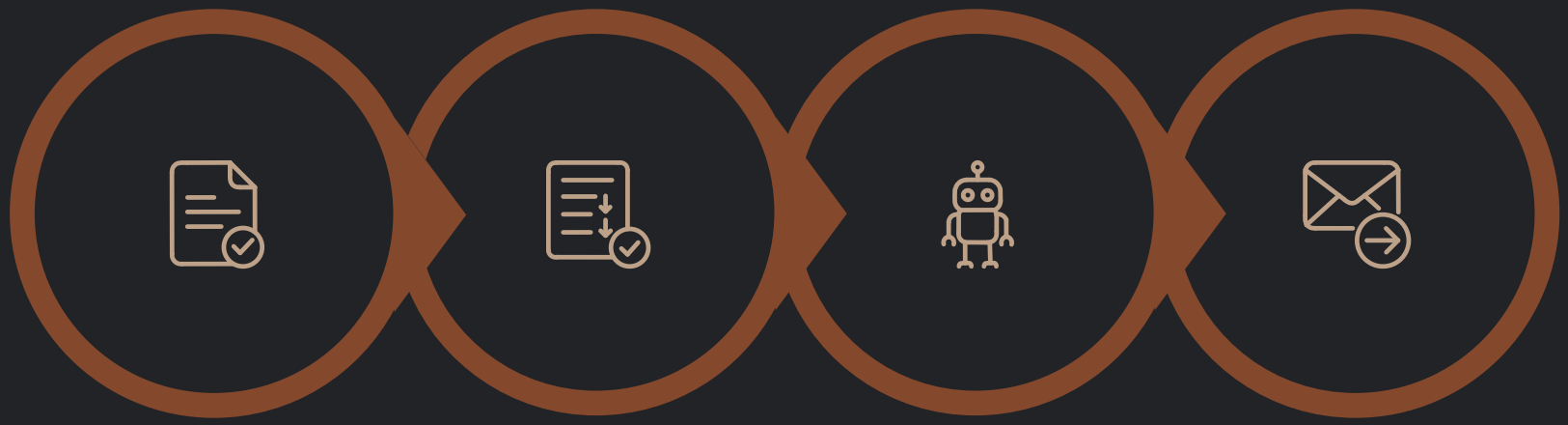
Workflow: Automating SOPs with Power Automate



Form Submit Trigger Flow AI Draft SOP Save & Email

Within minutes, every new project gets a standardized SOP covering Site Organisation, HR, EHS, and Site Closure — from **Day 0**, with zero copy-pasting.

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The Interface: Chatting with Your Manuals

What is RAG?

Retrieval-Augmented Generation — "Search first, then Answer." The AI only responds using your private company files.

> Example: "What are the key steps for Site Closure?" returns cited, verified steps — not a guess.

- Final measurements and bill settlements
- Release of securities and retentions
- Subcontractor settlements and cost-to-complete checks

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Safety First: The Human-in-the-Loop

In construction, a wrong answer can be dangerous. The system includes a strict **Escalation Protocol**.

LOW_CONFIDENCE Signal

If a question isn't covered by internal documents, the AI stops — it never guesses.

Automatic Escalation

Power Automate blocks the response and creates a high-priority ticket for a human manager to review.

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Why This Matters: The Board-Level Case

80%

Time Savings

Reduction in manual procedure drafting

0

Knowledge Loss

Senior engineers retire — their wisdom stays in the Digital Brain

100%

Source-Linked

Every answer cites the Board-Approved file and page

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Key Terms: Your Cheat Sheet

SOP	Standard Operating Procedure	The official "playbook" for a task
RAG	Retrieval-Augmented Generation	"Search first, then Answer" — AI uses only your facts
LLM	Large Language Model	The underlying "brain" (e.g. GPT-4) powering the AI
Metadata	Metadata	Labels on a file (Date, Author, Project Name) for easy retrieval
EHS	Environment, Health & Safety	Rules for keeping workers safe and sites compliant
API	Application Programming Interface	A digital "bridge" letting two apps talk to each other

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From Information Overload to Operational Excellence

> The future of construction isn't just about faster cranes or stronger concrete — it's about **smarter information**.

By creating a Digital Brain, no lesson is ever forgotten and every worker carries the wisdom of the entire company in their pocket.

Unified Knowledge

SharePoint as your corporate memory

Automated Workflows

Power Automate as your digital courier

Safe AI Answers

RAG + Human-in-the-Loop guardrails

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